CONTACTLESS DOORBELL USING ARDUINO

PRESENTED By :

PRASAD R S {22AI037}
NAGAJOTHI T R {22IT024}
GOVARTHINI M {22IT009}

Guided By :
MRS.LALITHA K

TEAM NO 24

22AIP10 & INTERNET OF THINGS LABORATORY

Abstract

1.A contactless doorbell system using Arduino and an ultrasonic sensor.

2.Detects the presence of a person near the door without physical touch.

3. When a person approaches, the ultrasonic sensor signals the Arduino.

4. Arduino triggers a buzzer or relay-controlled doorbell to sound.

5. Provides a hygienic solution, reducing the risk of germ transmission.

6. Can be integrated with smart home systems for notifications or camera activation.

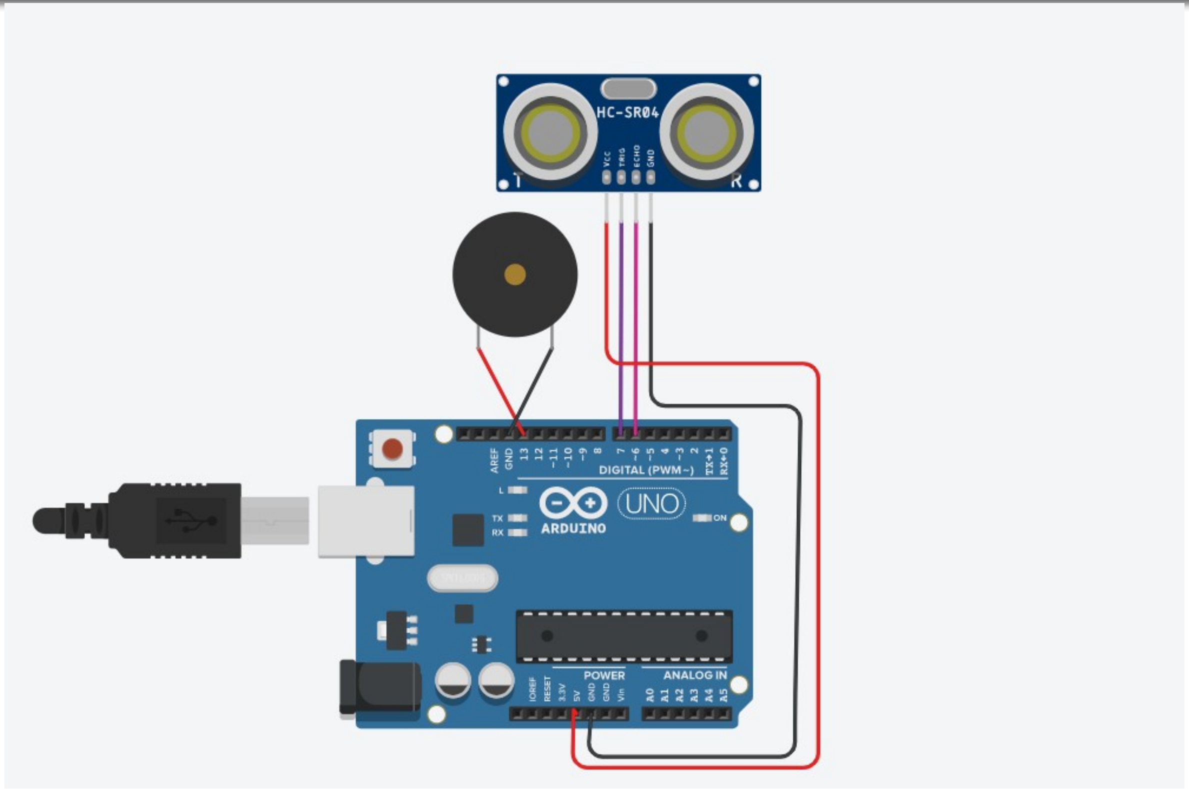
Objectives:

1.Design a contactless doorbell system using Arduino and an ultrasonic sensor.
2. Detect a person's presence and trigger a sound alert without touch.
3. Improve hygiene by eliminating physical contact with the doorbell.
-4.Create an efficient and scalable IoT solution for home automation.
-5.Enable integration with smart home systems for added functionality.

Components Required:

1. Arduino board (Uno, Nano, etc.)
2. Ultrasonic sensor (HC-SR04)
3. Buzzer or Speaker
4. Relay module (optional)
5. LEDs (optional)
6. Jumper wires
7. Breadboard
8. Power supply or USB cable

Flow Diagram:



Implementation:
(Hardware or software)

1. **Connect the Ultrasonic Sensor** : to the Arduino using jumper wires, ensuring proper connections for VCC, GND, Trig, and Echo pins.
2. **Wire the Buzzer :** or relay module to the Arduino to act as the doorbell alert mechanism.
3. **Upload the Arduino Code :** that reads distance data from the ultrasonic sensor and triggers the buzzer or relay when a person is detected within the set range.
4. **Test the System:** to ensure the sensor accurately detects proximity and the buzzer activates as expected.
5**. Optional :** Integrate LEDs for visual feedback or additional smart home features, such as notifications or camera activation.
6. **Assemble and Mount :** the components in a suitable enclosure and install the system near the door for operation.

Thank you